Hybridization and Sigma and Pi Bonds

Answer key with explanations

A. Hybridization

1. Hybridization.

> Hybridization is the mixing of an atom's valence orbitals into an equal number of identical hybrid orbitals. Carbon's 2s and 2p orbitals are different and the p orbitals are at 90°, yet methane has **four identical bonds at 109.5°**. Mixing one s and three p orbitals gives four identical **sp^3** hybrids pointing to the corners of a tetrahedron.

2. Hybridization and angles.

> (a) 4 domains: **sp^3, 109.5°** (b) 3 domains, no lone pair: **sp^2, 120°** (c) 2 domains: **sp, 180°** (d) 3 bonds + 1 lone pair: **sp^3**, about 107° (ideal 109.5°) (e) 3 domains: **sp^2, 120°** (f) 2 domains: **sp, 180°**

B. Sigma and pi bonds

3. σ and π.

> A **σ bond** forms by **head-on** overlap of orbitals along the line joining the nuclei; it allows rotation. A **π bond** forms by **side-on** overlap of two parallel p orbitals, with electron density above and below the line; it is weaker and prevents rotation.

4. Counting.

> (a) 6 C–H + 1 C–C: **7 σ, 0 π** (b) 4 C–H + C=C: **5 σ, 1 π** (c) N≡N: **1 σ, 2 π** (d) 6 C–H + C=C + C–C: **8 σ, 1 π** (e) two C=O: **2 σ, 2 π**

C. Carbon in organic molecules

5. Propyne.

> C1 (≡CH): 2 domains, **sp**, 180°. C2 (–C≡): 2 domains, **sp**, 180°. C3 (CH_3): 4 domains, **sp^3**, 109.5°. So the H–C≡C–C chain is linear.

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6. Isomers.

> The C=C contains a **π bond**, so rotation about it is not possible: the two Cl atoms can be fixed on the **same side** (*cis*) or **opposite sides** (*trans*), giving two different compounds. In 1,2-dichloroethane the C–C is only a **σ bond**, which rotates freely, so all arrangements interconvert and there is only one compound.

D. Lone pairs

7. Water and CO_2.

> Oxygen in water has 2 bonds **and 2 lone pairs**: 4 domains, sp^3, so the molecule is **bent** (about 104.5°). Carbon in CO_2 has 2 double bonds and **no lone pairs**: 2 domains, sp, so the molecule is **linear** (180°).

E. Challenge

8. Methanamide.

> (a) C has 3 domains (H, O, N): **sp^2**. O has 3 domains (the C and 2 lone pairs): **sp^2**. (b) Bonds: C–H, C=O, C–N, 2 × N–H: **5 σ, 1 π**. (c) About **120**° (trigonal planar around C).

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